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EU AI Act

MDR versus EU AI Act

A Mapping Exercise

August 2025

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EU AI Act vs MDR





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AI & Partners defends and extends the digital rights of users at risk around the world. By combining direct technical support, comprehensive policy engagement, global advocacy, grassroots professional services, regulatory interventions, and participating in industry groups such as AI Commons, we fight for fundamental rights in the artificial intelligence age.

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This report is an AI & Partners publication.





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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We’ve seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it’s handled responsibly. That’s why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

What Do We Do?

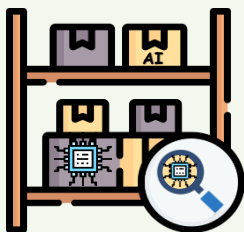
We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.



How Do We Do It?

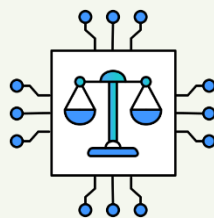
Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building ‘Trustworthy-by-Design’.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.



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Introduction

As the regulatory landscape evolves to keep pace with technological innovation, the intersection between the EU Medical Device Regulation (MDR) and the EU Artificial Intelligence Act (AI Act) is becoming increasingly important—especially for manufacturers of AI-enabled medical devices. While MDR governs the safety, performance, and post-market oversight of medical technologies, the AI Act introduces a complementary layer of regulation focused on the trustworthy development, deployment, and monitoring of high-risk AI systems.

This report serves as a mapping document to help stakeholders understand how the requirements of MDR align with the obligations introduced by the EU AI Act. It identifies areas of overlap—such as risk management, human oversight, quality systems, and post-market monitoring—where compliance efforts can be streamlined or reinforced to meet both sets of expectations.

By clarifying the touchpoints between MDR and the AI Act, this document aims to support a harmonized approach to compliance, ensuring that AI-enabled medical devices not only meet technical and clinical safety standards but also uphold principles of transparency, accountability, and robustness as required under the AI Act.

Whether you are a regulatory affairs professional, quality manager, product developer, or legal advisor, this report provides strategic guidance for integrating AI-specific governance into existing MDR frameworks—laying the foundation for secure, ethical, and compliant innovation in medical technologies.

Best regards,

Sean Musch

Founder/CEO

AI & Partners



Key questions being asked about MDR





1. What is the EU MDR and why was it introduced?

The EU MDR is the regulation that governs the safety, performance, and market access of medical devices in the European Union. It replaced the older Medical Device Directive (MDD) and Active Implantable Medical Device Directive (AIMDD) in May 2021. The MDR was introduced to address weaknesses exposed by past scandals (e.g., PIP breast implants), improve patient safety, and enhance device transparency. It includes stricter pre-market and post-market requirements, clearer definitions, greater oversight of Notified Bodies, and mandates the use of a central database (EUDAMED) to improve traceability and public access to information on devices and manufacturers.

2. What types of products are covered under the MDR?

MDR applies to medical devices for human use and their accessories, including implantable, active, and non-active devices. It also covers certain products without a medical purpose (e.g., colored contact lenses, aesthetic dermal fillers) listed in Annex XVI, due to their risk profile. Software with a medical purpose (e.g., diagnostic AI tools) is also considered a medical device. The MDR excludes in vitro diagnostic devices (regulated by IVDR), cosmetic products, and pharmaceuticals, unless the product has a combined function. Manufacturers must determine whether their product falls within scope and ensure it meets the relevant classification and regulatory requirements.

3. How does device classification work under MDR?

Under MDR, devices are classified into four risk-based categories: Class I (low risk), Class IIa and IIb (medium risk), and Class III (high risk). Classification is based on 22 rules in Annex VIII, which assess factors like invasiveness, duration of contact with the body, and whether the device is implantable or active. Software is classified based on how it influences clinical decisions. The higher the risk, the greater the regulatory scrutiny, and the more involvement is required from a Notified Body. Proper classification is essential as it determines the route of conformity assessment and documentation required for market approval.

4. What are the main obligations for manufacturers under MDR?

Manufacturers must ensure their devices comply with the General Safety and Performance Requirements (GSPRs) in Annex I, maintain a quality management system (QMS), and conduct clinical evaluations with sufficient data. They must register economic operators and devices in EUDAMED, assign UDI codes, and maintain post-market surveillance systems. Additionally, a Person Responsible for Regulatory Compliance (PRRC) must be designated. They are also required to report serious incidents and field safety corrective actions. Technical documentation must be prepared and kept up to date. These obligations aim to ensure product quality, traceability, and safety throughout the device lifecycle.

5. What is the Unique Device Identification (UDI) system?

The UDI system is a key traceability mechanism under MDR designed to uniquely identify and track medical devices throughout their lifecycle. Each device is assigned a UDI-DI (device identifier) and UDI-PI (production identifier). The UDI must appear on product labels and be registered in EUDAMED. It enhances transparency, facilitates recalls, improves inventory and incident management, and helps detect falsified devices. The system also supports post-market surveillance by linking reported incidents to specific devices. Class I devices have longer transition deadlines, while higher-risk devices must comply sooner. Software must display UDI in the interface if applicable.





6. What is EUDAMED and how does it impact compliance?

EUDAMED is the European database for medical devices, designed to centralize and enhance access to regulatory information. It integrates multiple modules, including those for device registration, clinical investigations, vigilance, market surveillance, certificates, and economic operators. Manufacturers must register device and organization information in EUDAMED, upload UDI data, and use it to report incidents and updates. For AI-enabled devices, it may eventually integrate with the EU AI Act database. EUDAMED promotes transparency, enabling healthcare professionals and the public to verify device status and safety. Compliance requires familiarity with its functions and timely data submission for all applicable products.

7. How does the MDR affect software and AI-based medical devices?

Software with a medical purpose is regulated under MDR and often classified as Class IIa or higher due to its influence on clinical decisions. The MDR recognizes software as an active device and sets strict rules for its evaluation. AI systems must demonstrate consistent performance, explainability, and reliability. With the upcoming EU AI Act, high-risk AI medical devices will also need to meet additional requirements around data quality, human oversight, and algorithm transparency. Companies must align MDR technical documentation with AI Act obligations, especially for post-market monitoring, update control, and risk mitigation strategies involving software behavior.

8. What are the clinical evaluation requirements under MDR?

Clinical evaluation is the ongoing process of assessing a device's safety and performance based on clinical data. MDR requires a more robust and transparent evaluation, especially for higher-risk devices and software. Clinical evidence must support all claims, and evaluations should reflect real-world use. For Class III and implantable devices, clinical investigations are generally mandatory. Manufacturers must document this in a Clinical Evaluation Report (CER), referencing peer-reviewed literature, existing clinical data, or new studies. Clinical data must be updated post-market through Post-Market Clinical Follow-Up (PMCF) activities, ensuring continuous validation of device effectiveness and safety.

9. What is the role of the Person Responsible for Regulatory Compliance (PRRC)?

The PRRC ensures that the manufacturer complies with MDR obligations. This includes oversight of device conformity, technical documentation, post-market surveillance, and incident reporting. The PRRC must possess specific qualifications: a degree in law, medicine, pharmacy, engineering, or a related field, and at least one year of regulatory experience, or four years of relevant experience without a degree. The PRRC must be permanently and continuously available within the organization or via contractual arrangements. The role reflects the MDR's emphasis on accountability and competence in regulatory compliance. Importers and authorized representatives must also designate a PRRC under certain conditions.

10. How does post-market surveillance work under MDR?

MDR requires all manufacturers to implement a Post-Market Surveillance (PMS) system that collects and analyzes data on device performance and safety after market entry. This includes user feedback, incident reports, complaints, and any trend data. For higher-risk devices, a Periodic Safety Update Report (PSUR) must be submitted regularly. PMS activities are part of the quality management system and should inform updates to technical documentation, labeling, and clinical evaluations. The goal is early identification of potential risks, continuous improvement, and swift action when safety concerns arise. Manufacturers must also report serious incidents and corrective actions to national authorities.



Understanding MDR





Scope and Definitions

Defines scope, products,
and terminology



What are the key
goals?

To define the scope of MDR and AI Act applicability to ensure AI-enabled medical devices are correctly regulated. It identifies whether a product is a medical device, an accessory, or a non-medical product with medical-like risks.



Why is it
needed?

Misclassification can lead to regulatory gaps, potentially compromising safety and performance. Clear definitions align AI-enabled medical devices with proper oversight under both MDR and the AI Act.



How does it work
in practice?

Enterprises must analyze the intended purpose and technology (e.g., machine learning diagnostics) to determine if their product is a medical device under MDR and a high-risk AI system under the AI Act.





Making Available, and Putting Into Service

Defines scope, products,
and terminology



What are the key
goals?

To ensure only compliant, safe, and performance-validated AI medical devices are placed on the EU market or put into service.



Why is it
needed?

Without strict entry conditions, patients and users may be exposed to unverified or biased AI algorithms. Both MDR and AI Act require documented assurance before deployment.



How does it work
in practice?

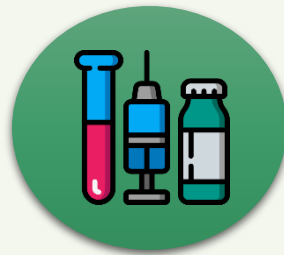
Manufacturers must CE-mark the device after demonstrating compliance with MDR's general safety and performance requirements and the AI Act's transparency, data governance, and risk management rules.





Identification and Traceability of Devices

UDI system ensures device traceability.



What are the key goals?

To improve post-market oversight, device recall efficiency, and traceability through unique identifiers and centralized databases.



Why is it needed?

AI systems can evolve post-deployment. Ensuring traceability is critical for tracking updates, version control, and safety incidents involving algorithm changes.

How does it work in practice?

Devices are labeled with Unique Device Identifiers (UDI), and manufacturers must register product data in EUDAMED. The AI Act may introduce a parallel registry for high-risk AI systems, requiring aligned record-keeping.





Notified Bodies

Oversight of Conformity Assessment Bodies



What are the key goals?

To designate competent authorities that assess conformity of AI-enabled medical devices against both MDR and AI Act requirements.



Why is it needed?

Assessing AI-specific risks (e.g., bias, explainability, robustness) requires specialized evaluators. Notified Bodies must understand how AI affects device behavior and safety.

How does it work in practice?

Manufacturers must choose Notified Bodies accredited for MDR. For high-risk AI systems, these bodies should have AI-specific expertise or collaborate with AI specialists to assess compliance with AI Act obligations.





Classification and Conformity Assessment

D Risk-based classification, conformity procedures.



What are the key goals?

To ensure devices are correctly classified by risk and assessed using the appropriate level of regulatory scrutiny.



Why is it needed?

AI can significantly increase risk (e.g., misdiagnosis), so risk-based classification and rigorous evaluation prevent unsafe devices from reaching the market.

How does it work in practice?

Devices are classified (I, IIa, IIb, III) under MDR based on invasiveness and risk. For AI systems, high-risk classification under the AI Act triggers additional conformity checks, including technical documentation review and post-market monitoring plans.





Clinical Evaluation and Assessments

Clinical evidence for safety, efficacy



What are the key goals?

To verify the safety, performance, and clinical benefit of AI-enabled devices through empirical evidence and ethical testing.



Why is it needed?

AI systems can underperform or exhibit bias if not properly validated across populations and clinical settings. Evaluation ensures real-world relevance.

How does it work in practice?

Manufacturers conduct clinical investigations in compliance with ISO 14155, documenting real-world AI performance, accuracy, and limitations. The AI Act mandates evaluation of training and test data quality as part of system validation.





Post-Market Surveillance, Vigilance, & Market Surveillance

Monitoring, incident reporting, regulatory
enforcement



What are the key
goals?

To monitor device performance and safety after market entry and respond to emerging risks, especially those stemming from AI behaviour over time.



Why is it
needed?

AI systems may change (e.g., adaptive learning) or behave unpredictably post-deployment. Surveillance ensures continued compliance and patient safety.

How does it work
in practice?

Enterprises must maintain a post-market surveillance (PMS) system that includes AI drift detection, update logging, and incident reporting to regulators. AI-specific incidents must also be submitted under AI Act obligations for high-risk systems.





Cooperation, Coordination, and Governance

EU coordination, expert group governance



What are the key
goals?

To harmonize MDR and AI Act enforcement across EU Member States via centralized coordination and expert groups.



Why is it
needed?

Different interpretations of AI compliance can fragment the market. Unified guidance supports legal certainty and regulatory predictability for AI medical device developers.

How does it work
in practice?

Manufacturers monitor guidance from the Medical Device Coordination Group (MDCG) and collaborate with national authorities. The AI Act introduces national AI supervisory bodies that must coordinate with MDR regulators.





Confidentiality, Data Protection, and Funding

Information protection and regulatory funding



What are the key goals?

To protect patient and user data and ensure regulatory transparency without compromising innovation or personal rights.



Why is it needed?

AI relies on large datasets, increasing the risk of privacy violations or data misuse. GDPR, MDR, and the AI Act intersect to safeguard personal data.

How does it work in practice?

AI developers must align their systems with GDPR, ensure anonymization or lawful data processing, and transparently document data practices. Regulatory fees and support systems must be disclosed and fairly applied.



Mapping MDR to EU AI Act

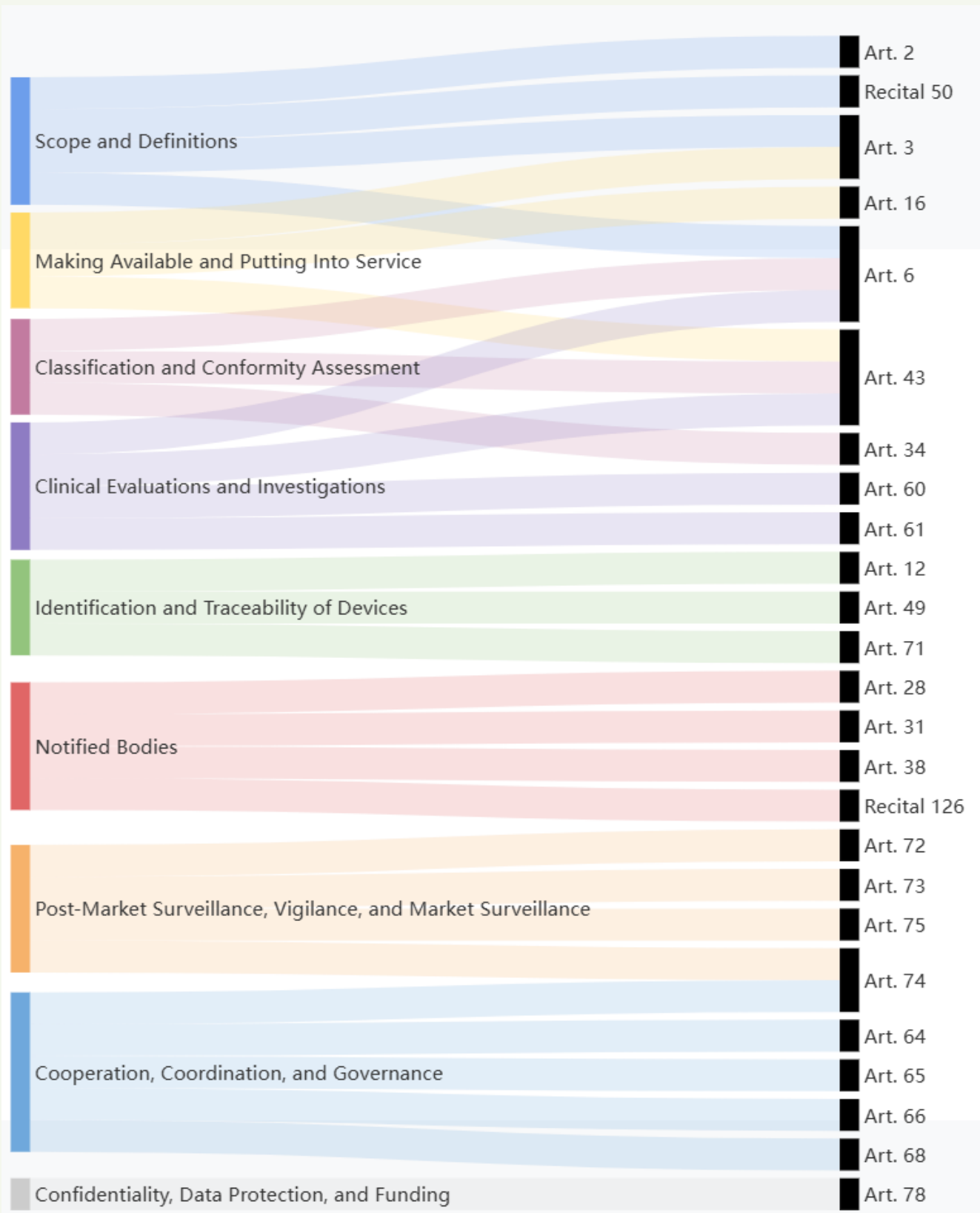




MDR

EU AI Act

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DORA		EU AI Act		
Section	Description	Provision(s)	Explanation	Action
Scope and Definitions	Chapter I sets out the scope and definitions of the Regulation. It clarifies which products are covered—primarily medical devices and their accessories—and explicitly excludes others such as in vitro diagnostics and certain medicinal products. It also introduces key definitions, including “medical device,” “accessory,” “UDI,” and “implantable device,” which underpin the rest of the Regulation.	2, 3, 6, Recital 50	Scope Scope of the Regulation: - The EU AI Act applies to providers and deployers of AI systems within the Union, regardless of whether they are established in the Union or a third country. It also covers importers, distributors, and manufacturers placing AI systems on the market. - The Act explicitly excludes AI systems used exclusively for military, defense, or national security purposes, and those used by public authorities in third countries under certain conditions. Exclusions: - The Act does not apply to AI systems developed for scientific research and development, nor to those used in personal non-professional activities. Definitions Key Definitions: AI System: Defined as a machine-based system designed to operate with varying levels of	◆ Assess whether your AI-enabled medical product falls under the EU AI Act and MDR. Review both MDR and AI Act definitions to determine if your device qualifies as a medical device with an embedded high-risk AI system. If so, both regulations apply concurrently, requiring dual compliance. + For high-risk AI systems, consider embedding auditability and explainability features (e.g., SHAP, LIME, counterfactuals) and conducting ethical assessments like FRIAs to ensure compliance and societal trust, as recommended by ISACA and AIGP.





			autonomy, capable of generating outputs such as predictions or decisions. • Provider: A person or entity that develops or has an AI system developed and places it on the market. • Deployer: A person or entity using an AI system under their authority. Specific Terms: • The Act does not specifically define terms like "medical device," "accessory," "UDI," or "implantable device" as these are more relevant to medical device regulations rather than AI systems. However, it does classify AI systems as high-risk if they are safety components of products covered by Union harmonization legislation, which includes medical devices.	
Making Available and Putting Into Service	Chapter II governs the placing of devices on the market and their availability and use. It outlines the essential conditions under	3, 16, 43	Placing on the Market and Putting Into Service Placing on the Market: • The EU AI Act defines "placing on the market" as the first making available of an AI system or a	◆ Verify CE marking and AI compliance before market launch. Ensure the AI system meets MDR safety and performance requirements and AI Act obligations for high-risk systems, such as





	which devices can be marketed in the EU and emphasizes the requirement for CE marking as proof of conformity with safety and performance requirements.		general-purpose AI model on the Union market. This involves ensuring that the AI system complies with the necessary requirements before it is made available for distribution or use. Putting Into Service: "Putting into service" refers to the supply of an AI system for first use directly to the deployer or for own use in the Union for its intended purpose. This step requires that the AI system is ready for its intended use and complies with all relevant regulations. Essential Conditions and CE Marking Conformity and CE Marking: - High-risk AI systems must undergo a conformity assessment to ensure they meet the requirements set out in the EU AI Act. This includes affixing the CE marking to indicate conformity with the regulation. The CE marking serves as proof that the AI system	risk management, data governance, and human oversight, before affixing the CE mark.
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			complies with safety and performance standards. Conformity Assessment: The conformity assessment for high-risk AI systems can involve internal control or the involvement of a notified body, depending on the availability and application of harmonized standards or common specifications. This process ensures that the AI system is safe and performs as intended before being placed on the market or put into service.	
Identification and Traceability of Devices	Chapter III introduces a comprehensive system for the identification and traceability of medical devices. Central to this is the Unique Device Identification (UDI) system, which enhances traceability and post-market surveillance. This chapter also mandates the use of EUDAMED, the central European database for medical devices, to manage device	12, 49, 71	Identification and Traceability Unique Device Identification (UDI) System: While the EU AI Act does not specifically mention a UDI system for AI systems, it emphasizes the importance of traceability for high-risk AI systems. This is achieved through the requirement for automatic recording of events (logs) over the system's lifetime to ensure	◆ Register the AI system in EUDAMED and prepare for future AI databases. Assign a UDI to the device and prepare for reporting AI-specific information to both EUDAMED and the upcoming EU AI database, as required under the AI Act for transparency and traceability.





	registration and monitoring data.		traceability and facilitate post-market monitoring. Registration and Monitoring: - The EU AI Act mandates the registration of high-risk AI systems in an EU database before they are placed on the market or put into service. This database is intended to contain information necessary for the identification and traceability of AI systems, similar to the role of EUDAMED for medical devices. EU Database for High-Risk AI Systems: - The EU database for high-risk AI systems is designed to store information about the systems, including the provider's details, the system's intended purpose, and its status (e.g., on the market, in service, recalled). This aligns with the function of EUDAMED in managing device registration and monitoring data.	
Notified Bodies	Chapter IV deals with notified bodies—	28, 31, 38, Recital 126	Notified Bodies Designation and Oversight:	♦ Work with a Notified Body with AI expertise.





	organizations designated to assess the conformity of medical devices. It sets strict criteria for their designation and ongoing oversight, requiring rigorous audits, including unannounced inspections. The chapter also highlights the role of expert panels in reviewing high-risk devices.		• The EU AI Act outlines the requirements for the designation and oversight of notified bodies responsible for assessing the conformity of high-risk AI systems. Notified bodies must be established under the national law of a Member State and have legal personality. • They must satisfy organizational, quality management, and resource requirements necessary to fulfill their tasks, ensuring independence and impartiality. Criteria for Designation: • Notified bodies must demonstrate competence, absence of conflicts of interest, and suitable cybersecurity measures. They are required to maintain confidentiality and have documented procedures to safeguard impartiality. Audits and Inspections: • Notified bodies are subject to regular	Ensure the Notified Body assessing your device is competent in evaluating AI components under both MDR and AI Act criteria, especially regarding algorithm transparency, data training integrity, and software updates.
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			monitoring and audits by notifying authorities to ensure compliance with the requirements. This includes the possibility of unannounced inspections to verify ongoing compliance. Role of Expert Panels: While the EU AI Act does not specifically mention expert panels for reviewing high-risk AI systems, it emphasizes the importance of coordination and cooperation among notified bodies and relevant stakeholders to ensure effective conformity assessments.	
Classification and Conformity Assessment	Chapter V addresses the classification of medical devices based on risk and details the conformity assessment procedures required for each class. Devices are categorized from Class I (low risk) to Class III (high risk), with increasing regulatory scrutiny. It also establishes	6, 34, 43	Classification of AI Systems Classification of High-Risk AI Systems: The EU AI Act classifies AI systems as high-risk based on their intended use and potential impact on health, safety, and fundamental rights. High-risk AI systems include those used as safety components in products covered	◆ Apply a dual conformity assessment approach. For AI-powered medical devices, conduct MDR risk classification and ensure compliance with AI Act obligations for high-risk AI systems, such as robustness, cybersecurity, and conformity procedures (e.g., internal checks or third-party audits).





	clinical evaluation requirements necessary for demonstrating conformity.		by Union harmonization legislation, such as medical devices. - The classification criteria are detailed in the Act, which specifies that AI systems are considered high-risk if they are safety components of products requiring third-party conformity assessment. Conformity Assessment Procedures Conformity Assessment for High-Risk AI Systems: - High-risk AI systems must undergo a conformity assessment to ensure compliance with the requirements set out in the EU AI Act. This process involves either internal control or the involvement of a notified body, depending on the application of harmonized standards or common specifications. - The conformity assessment procedures provide options for assessment based on the use of harmonized	
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			standards or common specifications. Role of Notified Bodies: - Notified bodies play a crucial role in the conformity assessment of high-risk AI systems. They are responsible for verifying compliance with the requirements and issuing certificates of conformity. - The operational obligations of notified bodies ensure they conduct assessments without imposing unnecessary burdens on providers. Clinical Evaluation Requirements: - While the EU AI Act does not specifically address clinical evaluation requirements for AI systems, it emphasizes the need for a comprehensive assessment of the system's impact on health, safety, and fundamental rights as part of the conformity assessment process.	
Clinical Evaluations and Investigations	Chapter VI focuses on clinical evaluation and clinical	6, 43, 60, 61	Clinical Evaluation and Investigations Conditions for Conducting Studies:	◆ Incorporate AI performance in clinical evaluation





	investigations. It lays out the conditions for conducting clinical studies in the EU, with attention to subject protection, data integrity, and ethical review. It mandates that clinical evidence be obtained to support the safety and performance claims of the device.		- The EU AI Act does not specifically address clinical evaluations or investigations as related to medical devices. However, it emphasizes the importance of testing high-risk AI systems in real-world conditions to ensure their safety and performance. This includes obtaining informed consent from subjects and ensuring data integrity and ethical oversight. Subject Protection: - The Act mandates that informed consent be obtained from subjects participating in real-world testing of high-risk AI systems. This consent must be freely given, specific, and informed, ensuring that subjects are aware of the nature, objectives, and potential inconveniences of the testing. Data Integrity and Ethical Review: - The Act requires that testing in real-world conditions be conducted with a	Include algorithm validation, bias analysis, and real-world performance monitoring in the clinical evaluation. Clearly demonstrate the reliability and safety of the AI under intended use conditions.
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			focus on data integrity and ethical considerations. Providers must ensure that the testing is overseen by qualified individuals and that personal data is handled in compliance with Union law. Supporting Safety and Performance Claims: While the Act does not specifically require clinical evidence for AI systems, it mandates that high-risk AI systems undergo a conformity assessment to demonstrate compliance with safety and performance requirements. This process ensures that the systems meet the necessary standards before being placed on the market or put into service.	
Post-Market Surveillance, Vigilance, and Market Surveillance	Chapter VII covers post-market surveillance and vigilance. Manufacturers are required to monitor the performance of devices after they are placed on the	72, 73, 74, 75	Post-Market Surveillance Post-Market Monitoring by Providers: The EU AI Act mandates that providers of high-risk AI systems establish a post-market monitoring system. This	◆ Establish AI-specific PMS and monitoring systems. Integrate continuous monitoring for algorithm drift, unintended consequences, and performance degradation. Report significant AI updates or





	market through robust post-market surveillance systems. The chapter also sets rules for reporting and investigating serious incidents and field safety corrective actions, and it outlines national market surveillance responsibilities.		system should be proportionate to the nature and risks of the AI system and actively collect, document, and analyze data on the system's performance throughout its lifetime. Post-Market Monitoring Plan: - Providers must develop a post-market monitoring plan as part of the technical documentation. This plan should include elements that allow for the evaluation of the AI system's continuous compliance with regulatory requirements. Vigilance and Reporting of Serious Incidents: - Providers of high-risk AI systems are required to report any serious incidents to the market surveillance authorities of the Member States where the incident occurred. The report must be made immediately after establishing a causal link between the AI system and the	safety issues under both MDR vigilance and AI Act incident reporting mechanisms.
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			incident, or within a specified timeframe. Investigation of Incidents: - Following the reporting of a serious incident, providers must conduct necessary investigations and cooperate with competent authorities and notified bodies during these investigations. Market Surveillance Market Surveillance Responsibilities: - Market surveillance authorities are responsible for ensuring compliance with the EU AI Act. They have the power to monitor, supervise, and enforce the regulation, including the ability to propose joint activities and investigations to promote compliance and identify non-compliance. Mutual Assistance and Information Sharing: - The Act provides for mutual assistance and information sharing among market surveillance authorities to	
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			ensure effective enforcement and control of AI systems in the Union market.	
Cooperation, Coordination, and Governance	Chapter VIII introduces governance mechanisms and cooperation between Member States. It establishes the Medical Device Coordination Group (MDCG), which supports harmonized implementation of the Regulation, and sets out procedures for expert panels and laboratories to assist with technical and clinical evaluations.	64, 65, 66, 68, 74	Governance Mechanisms European Artificial Intelligence Board: - The EU AI Act establishes the European Artificial Intelligence Board (the Board) to facilitate consistent and effective application of the Regulation across Member States. The Board coordinates among national competent authorities and supports joint activities of market surveillance authorities. AI Office: - The AI Office is tasked with developing Union expertise and capabilities in AI, supporting the Board, and facilitating cooperation among Member States. Cooperation and Coordination Coordination Among Authorities: - The Board contributes to the coordination among national competent	◆ Align with cross-sectoral oversight mechanisms. Track guidance from the Medical Device Coordination Group (MDCG) and AI Act supervisory bodies (e.g., national competent authorities or the European AI Board) to harmonize regulatory practices and anticipate enforcement trends.





			authorities and supports the harmonization of administrative practices, including the functioning of AI regulatory sandboxes and testing in real-world conditions. Expert Panels and Laboratories: - While the EU AI Act does not specifically mention expert panels and laboratories for technical and clinical evaluations, it does establish a scientific panel of independent experts to support enforcement activities and provide advice on AI systems. Market Surveillance and Joint Activities: - The Act allows for joint activities and investigations by market surveillance authorities to promote compliance and identify non-compliance across Member States.	
Confidentiality, Data Protection, and Funding	Chapter IX provides rules on confidentiality, personal data protection, and funding	78	Confidentiality Confidentiality Obligations: The EU AI Act mandates that the Commission,	◆ Embed GDPR and AI Act data principles into system design. Ensure your AI complies with data privacy requirements (GDPR),





	mechanisms. It ensures that sensitive information is protected and describes the responsibilities of the European Commission and Member States in supporting the Regulation, including transparency and fee structures.		market surveillance authorities, notified bodies, and other involved parties respect the confidentiality of information obtained during their activities. This includes protecting intellectual property rights, trade secrets, and other sensitive information. - Confidentiality is crucial for the effective implementation of the Regulation, particularly during inspections, investigations, or audits. Data Protection Data Protection Measures: - The Act requires that any data collected by authorities for assessing AI systems' risks be strictly necessary and protected by adequate cybersecurity measures. Data must be deleted once it is no longer needed for its original purpose. - The confidentiality obligations extend to	and that training and inference data used in AI development are properly anonymized, consented, and traceable.
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			information exchanged between national competent authorities and the Commission, especially when it involves high-risk AI systems used by law enforcement or other sensitive areas. Funding Mechanisms Support and Funding: • While the specific details on funding mechanisms are not explicitly outlined in the provided references, the EU AI Act involves the European Commission and Member States in supporting the Regulation's implementation. This includes ensuring transparency and possibly establishing fee structures to support regulatory activities.	
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Calls to action





Embed MDR into High-Risk AI System Monitoring

Under the EU AI Act, high-risk AI systems require ongoing post-deployment monitoring and robust incident management. MDR provides continuous threat detection, real-time triage, and expert response to ensure these systems operate securely and in compliance—even under attack.



Use MDR to Support AI Risk and Incident Reporting Obligations

Article 62 mandates serious incident and malfunction reporting for high-risk AI. MDR equips you with expert-led detection and contextual analysis to identify AI-related threats early, streamline reporting, and meet legal obligations without delay.



Integrate MDR to Strengthen AI System Cybersecurity-by-Design

Articles 15 and 17 demand cybersecurity “by design and by default” and resilient quality management systems. MDR complements these mandates by embedding threat intelligence, proactive defence, and attack surface management into your AI lifecycle—from development to deployment.



Leverage MDR to Maintain Human Oversight and Operational Control

The EU AI Act emphasizes human oversight for high-risk AI. MDR extends your oversight capabilities with 24/7 monitoring, expert-led analysis, and contextual threat intelligence—ensuring humans remain in control, even in complex or automated environments.





Conclusion

The intersection of the EU Medical Device Regulation (MDR) and the EU AI Act signals a critical shift in how AI-enabled medical technologies are governed—moving toward a more integrated, accountable, and risk-based regulatory model. As medical devices increasingly incorporate AI, aligning MDR’s safety and performance requirements with the AI Act’s focus on transparency, oversight, and lifecycle risk management is essential to ensuring both compliance and patient trust.

This mapping document has highlighted the complementary roles these frameworks play: MDR provides the foundation for device safety and post-market surveillance, while the AI Act introduces additional obligations specific to high-risk AI systems, including human oversight, data governance, and algorithmic robustness. For manufacturers, notified bodies, and regulators alike, understanding these intersections is key to building harmonized compliance strategies that avoid redundancy and support innovation.

However, successful alignment will depend on operational execution. Stakeholders must adapt quality management systems, technical documentation, and post-market plans to reflect the AI Act’s requirements—especially for AI systems classified as high-risk under Annex III. Smaller organizations may face additional challenges and will benefit from early planning, targeted guidance, and cross-functional collaboration between regulatory, technical, and clinical teams.

Despite these complexities, forward-looking organizations are already demonstrating that integrating MDR and AI Act compliance can enhance product quality, streamline market access, and strengthen long-term competitiveness. As AI becomes an increasingly central component of medical technologies, these frameworks offer a unified path toward safe, ethical, and effective AI-enabled care.





References

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